

Universidad Autónoma de Santo Domingo

Primada de América | Fundada el 28 de octubre de 1538



U03.LAB10

Ciencia de Datos II

Maestro Edwin R. José Nolasco



MAESTRANTE

MANUEL DE JESUS MAÑANÁ SANTANA

INF-8239-C2

INFORME DE ENTREGA DE PROYECTO: U3.LAB010

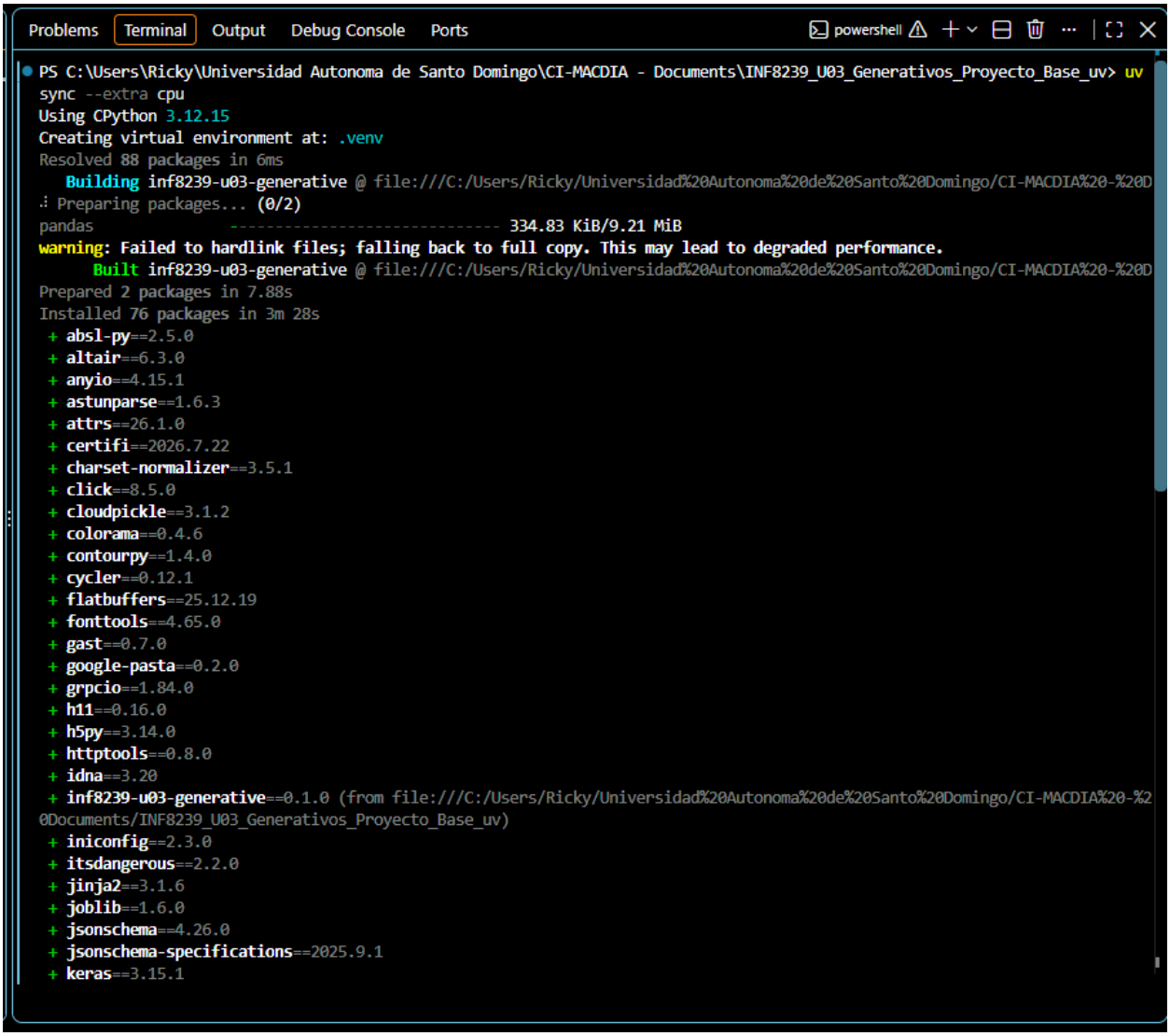
Autoencoder y VAE con Fashion-MNIST

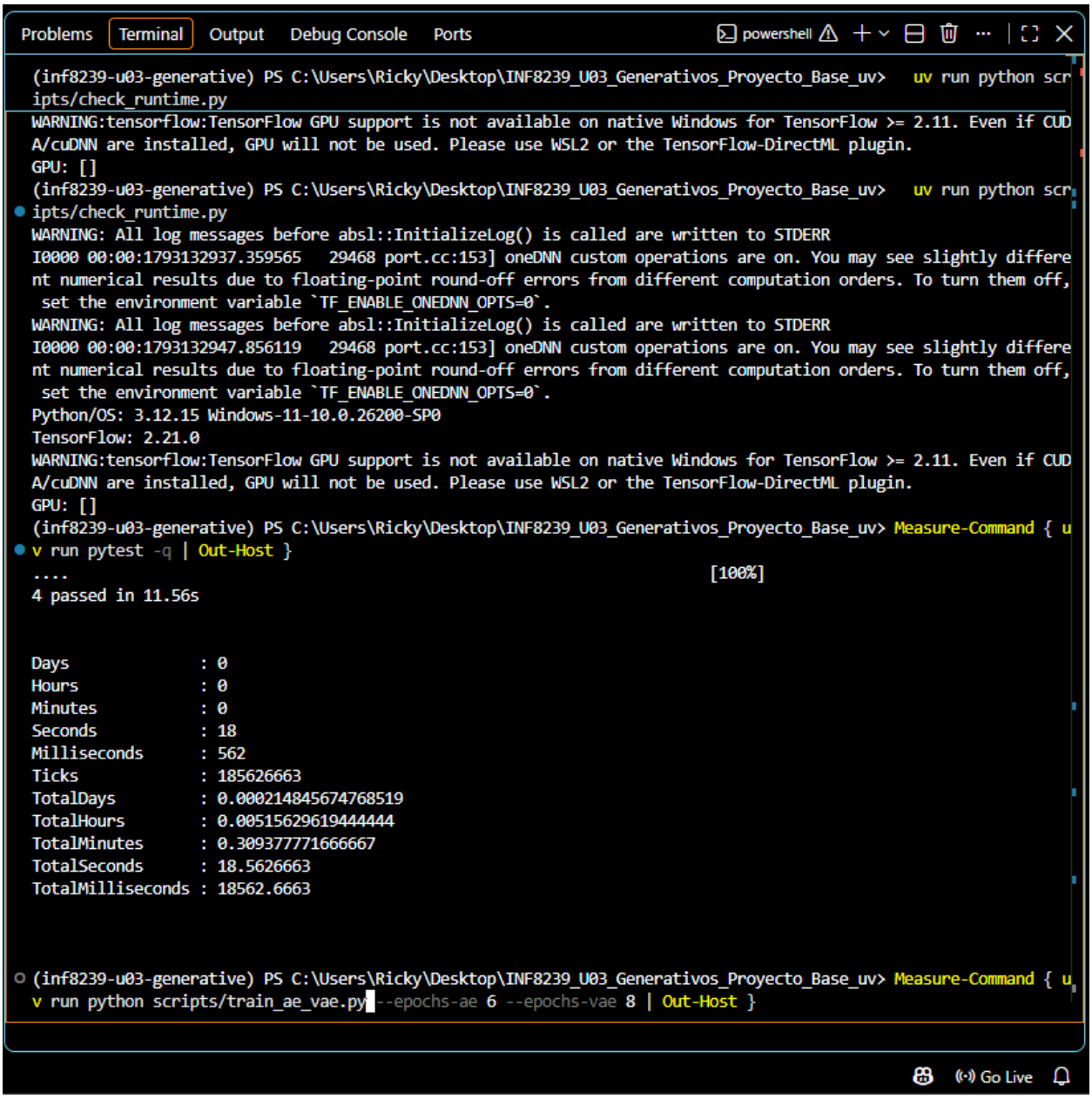
Reconstrucción, generación, espacio latente, costo y responsabilidad

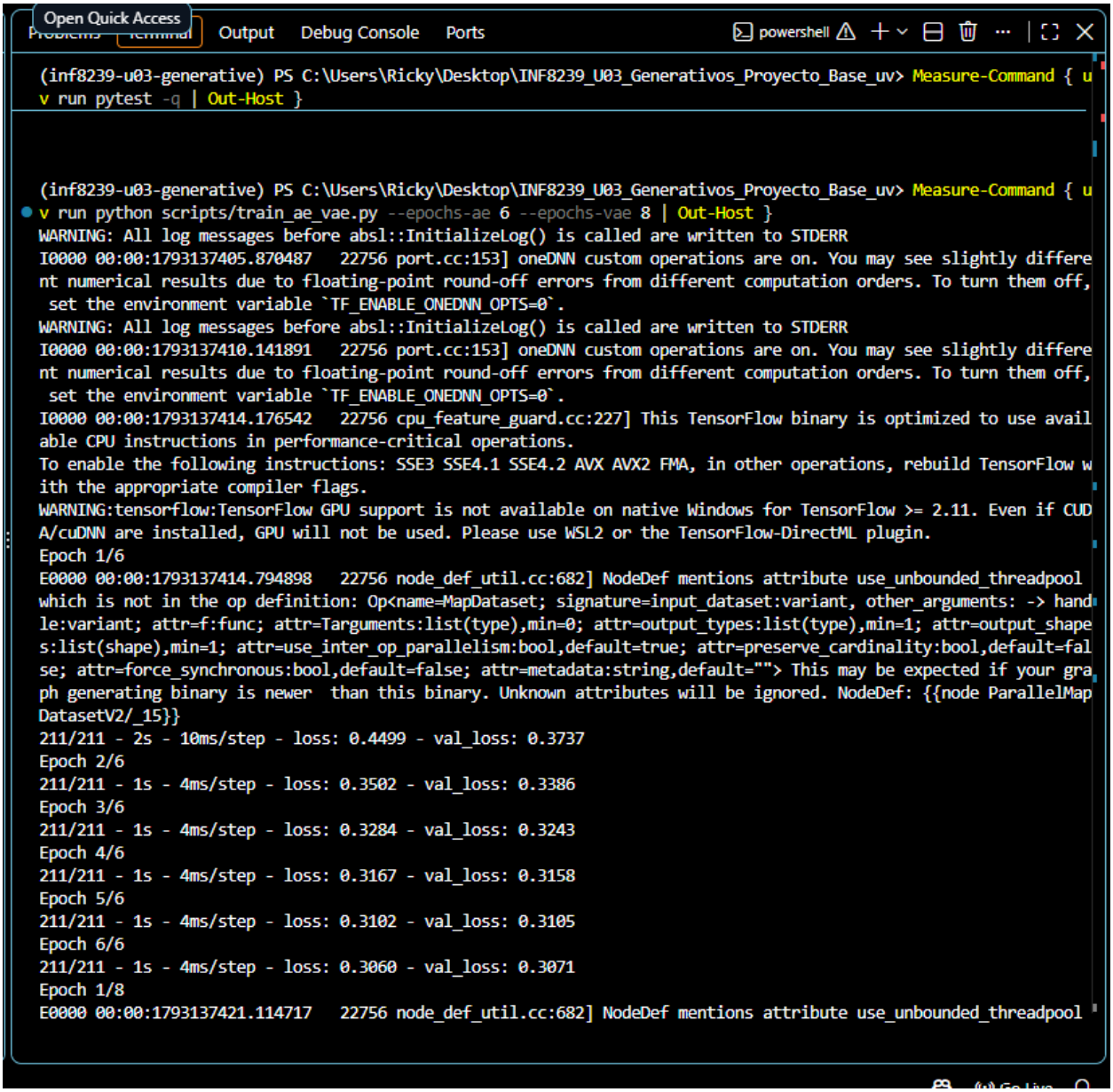
1. Enlace al Repositorio del Proyecto

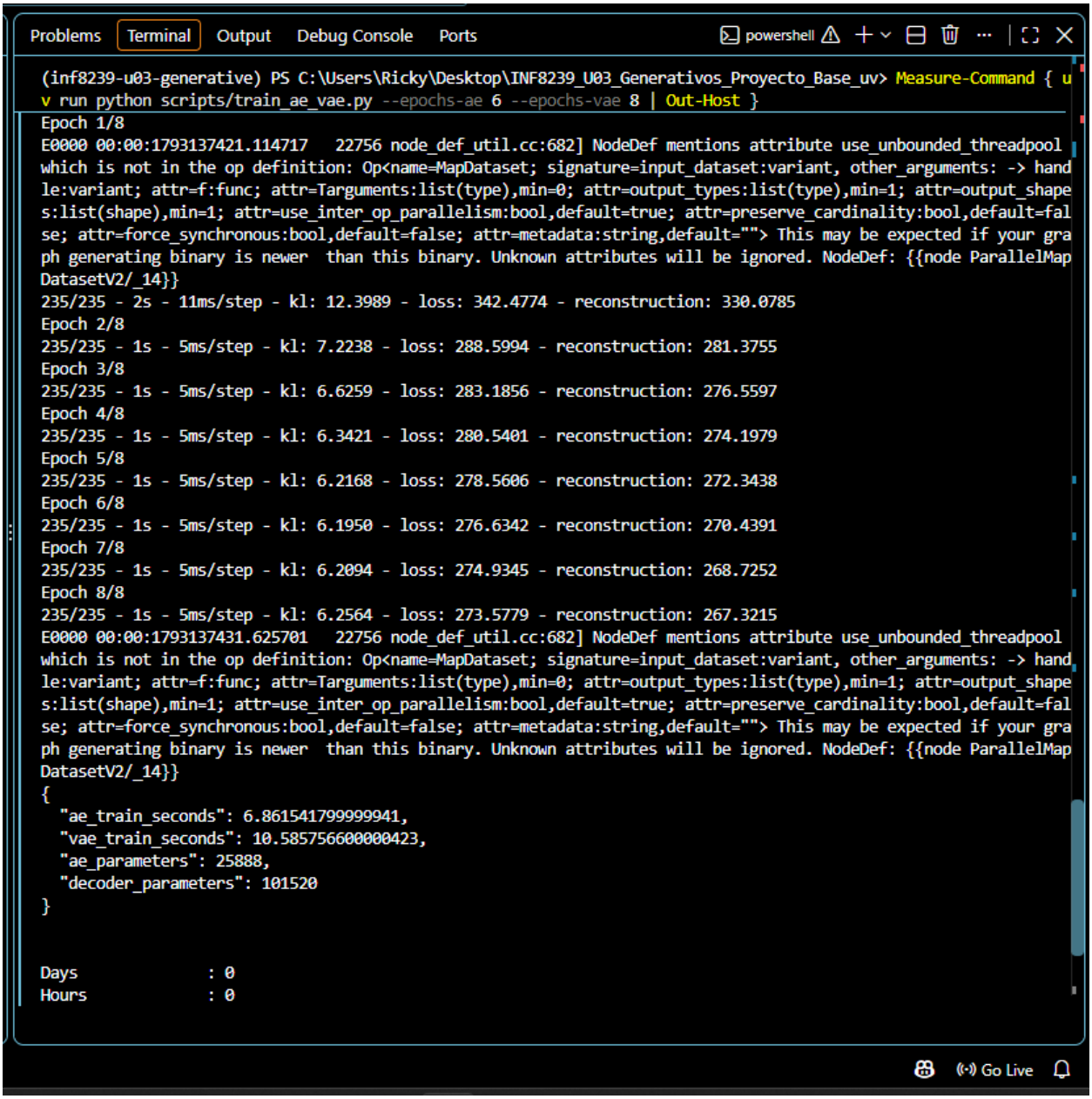
- **URL del Repositorio:** https://github.com/Rickymjms/INF8239_U03_Generativos_Proyecto_Base_uv.git
- **Rama Principal: main**
- **Visibilidad: Público**

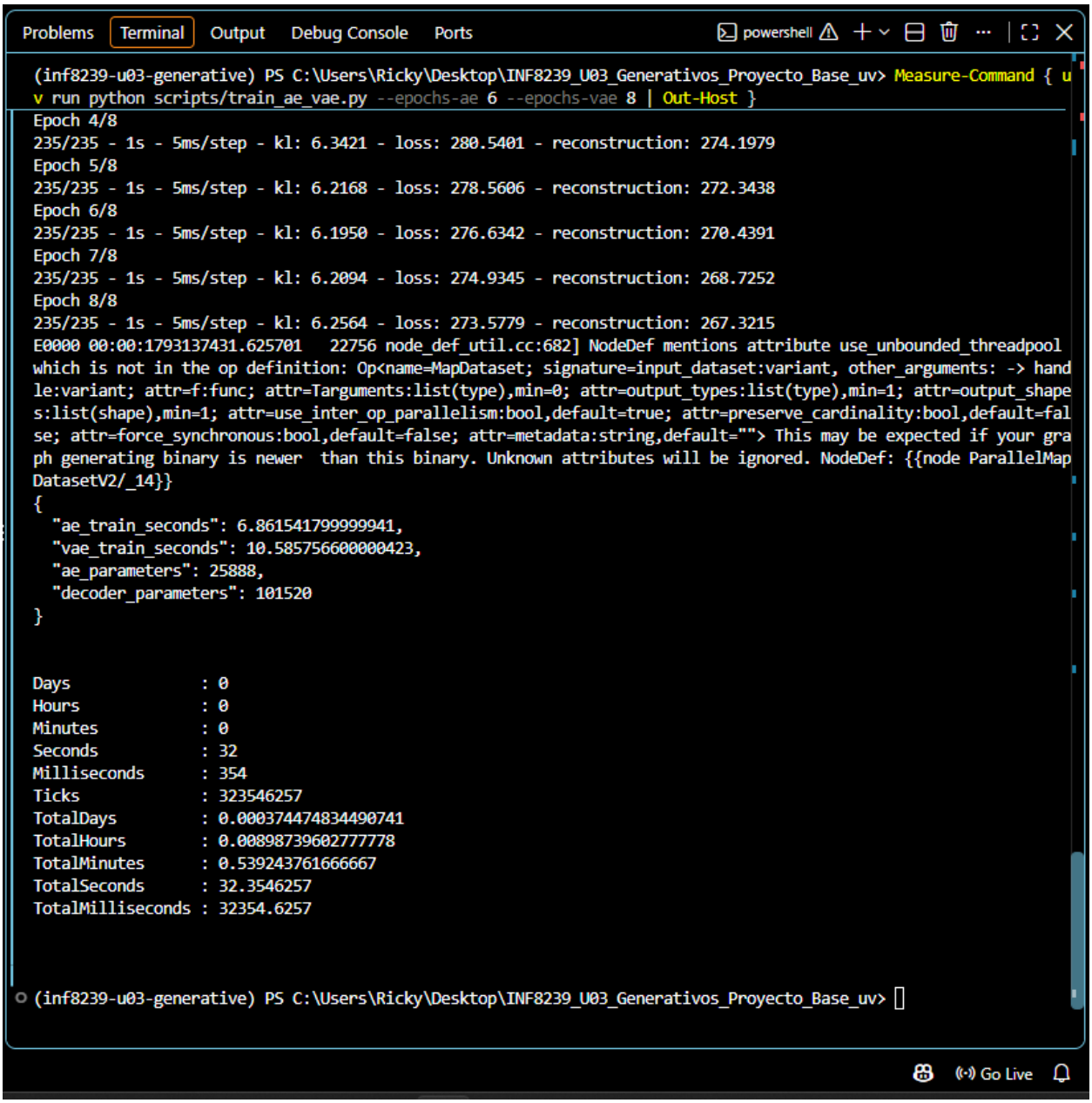
INF-8239 · Unidad 03 · Autoencoder y VAE con Fashion-MNIST

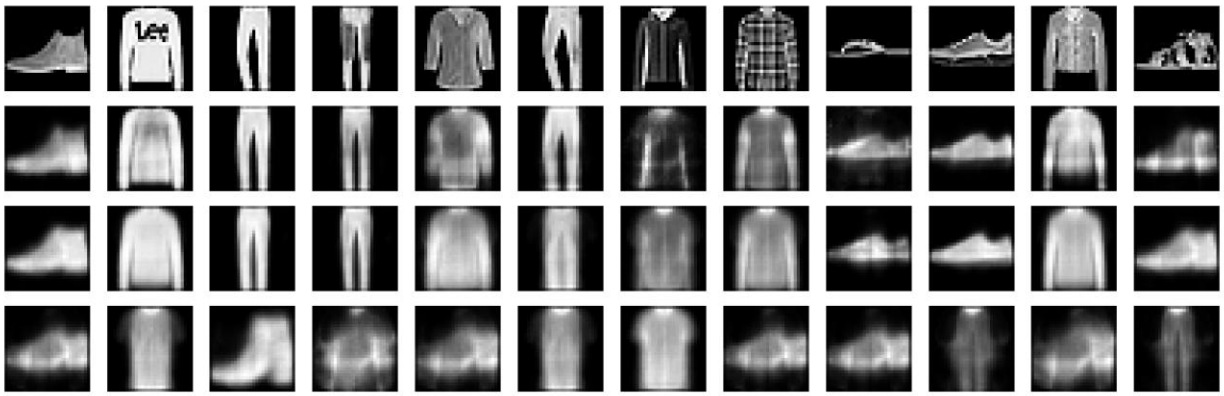


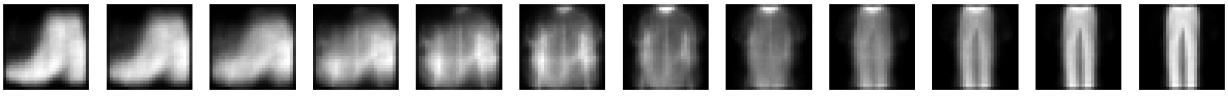

Problems
Terminal
Output
Debug Console
Ports
powershell
(inf8239-u03-generative) PS C:\Users\Ricky\Desktop\INF8239_U03_Generativos_Proyecto_Base_uv> uv run python scripts/check_runtime.py
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.
GPU: []
(inf8239-u03-generative) PS C:\Users\Ricky\Desktop\INF8239_U03_Generativos_Proyecto_Base_uv> uv run python scripts/check_runtime.py
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1793132937.359565 29468 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1793132947.856119 29468 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
Python/OS: 3.12.15 Windows-11-10.0.26200-SP0
TensorFlow: 2.21.0
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.
GPU: []
(inf8239-u03-generative) PS C:\Users\Ricky\Desktop\INF8239_U03_Generativos_Proyecto_Base_uv> Measure-Command { uv run pytest -q | Out-Host }
.... [100%]
4 passed in 11.56s
Days : 0
Hours : 0
Minutes : 0
Seconds : 18
Milliseconds : 562
Ticks : 185626663
TotalDays : 0.000214845674768519
TotalHours : 0.00515629619444444
TotalMinutes : 0.309377771666667
TotalSeconds : 18.5626663
TotalMilliseconds : 18562.6663
(inf8239-u03-generative) PS C:\Users\Ricky\Desktop\INF8239_U03_Generativos_Proyecto_Base_uv> Measure-Command { uv run python scripts/train_ae_vae.py --epochs-ae 6 --epochs-vae 8 | Out-Host }
Go Live


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Output
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powershell
(inf8239-u03-generative) PS C:\Users\Ricky\Desktop\INF8239_U03_Generativos_Proyecto_Base_uv> Measure-Command { uv run pytest -q | Out-Host }
(inf8239-u03-generative) PS C:\Users\Ricky\Desktop\INF8239_U03_Generativos_Proyecto_Base_uv> Measure-Command { uv run python scripts/train_ae_vae.py --epochs-ae 6 --epochs-vae 8 | Out-Host }
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1793137405.870487 22756 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1793137410.141891 22756 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
I0000 00:00:1793137414.176542 22756 cpu_feature_guard.cc:227] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.
Epoch 1/6
E0000 00:00:1793137414.794898 22756 node_def_util.cc:682] NodeDef mentions attribute use_unbounded_threadpool which is not in the op definition: Op<name=MapDataset; signature=input_dataset:variant, other_arguments: -> handle:variant; attr=f:func; attr=Targuments:list(type),min=0; attr=output_types:list(type),min=1; attr=output_shapes:list(shape),min=1; attr=use_inter_op_parallelism:bool,default=true; attr=preserve_cardinality:bool,default=false; attr=force_synchronous:bool,default=false; attr=metadata:string,default=""> This may be expected if your graph generating binary is newer than this binary. Unknown attributes will be ignored. NodeDef: {{node ParallelMapDatasetV2/_15}}
211/211 - 2s - 10ms/step - loss: 0.4499 - val_loss: 0.3737
Epoch 2/6
211/211 - 1s - 4ms/step - loss: 0.3502 - val_loss: 0.3386
Epoch 3/6
211/211 - 1s - 4ms/step - loss: 0.3284 - val_loss: 0.3243
Epoch 4/6
211/211 - 1s - 4ms/step - loss: 0.3167 - val_loss: 0.3158
Epoch 5/6
211/211 - 1s - 4ms/step - loss: 0.3102 - val_loss: 0.3105
Epoch 6/6
211/211 - 1s - 4ms/step - loss: 0.3060 - val_loss: 0.3071
Epoch 1/8
E0000 00:00:1793137421.114717 22756 node_def_util.cc:682] NodeDef mentions attribute use_unbounded_threadpool


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(inf8239-u03-generative) PS C:\Users\Ricky\Desktop\INF8239_U03_Generativos_Proyecto_Base_uv> Measure-Command { uv run python scripts/train_ae_vae.py --epochs-ae 6 --epochs-vae 8 | Out-Host }
Epoch 1/8
E0000 00:00:1793137421.114717 22756 node_def_util.cc:682] NodeDef mentions attribute use_unbounded_threadpool which is not in the op definition: Op<name=MapDataset; signature=input_dataset:variant, other_arguments: -> handle:variant; attr=f:func; attr=Targuments:list(type),min=0; attr=output_types:list(type),min=1; attr=output_shapes:list(shape),min=1; attr=use_inter_op_parallelism:bool,default=true; attr=preserve_cardinality:bool,default=false; attr=force_synchronous:bool,default=false; attr=metadata:string,default=""> This may be expected if your graph generating binary is newer than this binary. Unknown attributes will be ignored. NodeDef: {{node ParallelMapDatasetV2/_14}}
235/235 - 2s - 11ms/step - kl: 12.3989 - loss: 342.4774 - reconstruction: 330.0785
Epoch 2/8
235/235 - 1s - 5ms/step - kl: 7.2238 - loss: 288.5994 - reconstruction: 281.3755
Epoch 3/8
235/235 - 1s - 5ms/step - kl: 6.6259 - loss: 283.1856 - reconstruction: 276.5597
Epoch 4/8
235/235 - 1s - 5ms/step - kl: 6.3421 - loss: 280.5401 - reconstruction: 274.1979
Epoch 5/8
235/235 - 1s - 5ms/step - kl: 6.2168 - loss: 278.5606 - reconstruction: 272.3438
Epoch 6/8
235/235 - 1s - 5ms/step - kl: 6.1950 - loss: 276.6342 - reconstruction: 270.4391
Epoch 7/8
235/235 - 1s - 5ms/step - kl: 6.2094 - loss: 274.9345 - reconstruction: 268.7252
Epoch 8/8
235/235 - 1s - 5ms/step - kl: 6.2564 - loss: 273.5779 - reconstruction: 267.3215
E0000 00:00:1793137431.625701 22756 node_def_util.cc:682] NodeDef mentions attribute use_unbounded_threadpool which is not in the op definition: Op<name=MapDataset; signature=input_dataset:variant, other_arguments: -> handle:variant; attr=f:func; attr=Targuments:list(type),min=0; attr=output_types:list(type),min=1; attr=output_shapes:list(shape),min=1; attr=use_inter_op_parallelism:bool,default=true; attr=preserve_cardinality:bool,default=false; attr=force_synchronous:bool,default=false; attr=metadata:string,default=""> This may be expected if your graph generating binary is newer than this binary. Unknown attributes will be ignored. NodeDef: {{node ParallelMapDatasetV2/_14}}
{
"ae_train_seconds": 6.861541799999941,
"vae_train_seconds": 10.585756600000423,
"ae_parameters": 25888,
"decoder_parameters": 101520
}
Days : 0
Hours : 0
Go Live


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(inf8239-u03-generative) PS C:\Users\Ricky\Desktop\INF8239_U03_Generativos_Proyecto_Base_uv> Measure-Command { uv run python scripts/train_ae_vae.py --epochs-ae 6 --epochs-vae 8 | Out-Host }
Epoch 4/8
235/235 - 1s - 5ms/step - kl: 6.3421 - loss: 280.5401 - reconstruction: 274.1979
Epoch 5/8
235/235 - 1s - 5ms/step - kl: 6.2168 - loss: 278.5606 - reconstruction: 272.3438
Epoch 6/8
235/235 - 1s - 5ms/step - kl: 6.1950 - loss: 276.6342 - reconstruction: 270.4391
Epoch 7/8
235/235 - 1s - 5ms/step - kl: 6.2094 - loss: 274.9345 - reconstruction: 268.7252
Epoch 8/8
235/235 - 1s - 5ms/step - kl: 6.2564 - loss: 273.5779 - reconstruction: 267.3215
E0000 00:00:1793137431.625701 22756 node_def_util.cc:682] NodeDef mentions attribute use_unbounded_threadpool which is not in the op definition: Op<name=MapDataset; signature=input_dataset:variant, other_arguments: -> handle:variant; attr=f:func; attr=Targuments:list(type),min=0; attr=output_types:list(type),min=1; attr=output_shapes:list(shape),min=1; attr=use_inter_op_parallelism:bool,default=true; attr=preserve_cardinality:bool,default=false; attr=force_synchronous:bool,default=false; attr=metadata:string,default=""> This may be expected if your graph generating binary is newer than this binary. Unknown attributes will be ignored. NodeDef: {{node ParallelMapDatasetV2/_14}}
{
"ae_train_seconds": 6.861541799999941,
"vae_train_seconds": 10.585756600000423,
"ae_parameters": 25888,
"decoder_parameters": 101520
}
Days : 0
Hours : 0
Minutes : 0
Seconds : 32
Milliseconds : 354
Ticks : 323546257
TotalDays : 0.000374474834490741
TotalHours : 0.00898739602777778
TotalMinutes : 0.539243761666667
TotalSeconds : 32.3546257
TotalMilliseconds : 32354.6257
(inf8239-u03-generative) PS C:\Users\Ricky\Desktop\INF8239_U03_Generativos_Proyecto_Base_uv>
Go Live


Lee



INF-8239 · Explorador del espacio latente

Demostración académica con Fashion-MNIST. Una muestra sintética no garantiza originalidad ni privacidad.

Latente X

0.00

Latente Y

3.00



Punto latente (0.0, 3.0)

INF-8239 · Explorador del espacio latente

Demostración académica con Fashion-MNIST. Una muestra sintética no garantiza originalidad ni privacidad.

Latente X

0.00

Latente Y

-3.00



Punto latente (0.0, -3.0)

INF-8239 · Explorador del espacio latente

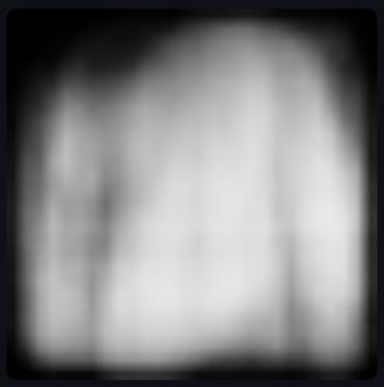
Demostración académica con Fashion-MNIST. Una muestra sintética no garantiza originalidad ni privacidad.

Latente X

-3.00

Latente Y

0.00



Punto latente (-3.0, 0.0)

INF-8239 · Explorador del espacio latente

Demostración académica con Fashion-MNIST. Una muestra sintética no garantiza originalidad ni privacidad.

Latente X

3.00

Latente Y

0.00



Punto latente (3.0, 0.0)

INF-8239 · Explorador del espacio latente

Demostración académica con Fashion-MNIST. Una muestra sintética no garantiza originalidad ni privacidad.

Latente X

1.50

Latente Y

0.00



Punto latente (1.5, 0.0)

INF-8239 · Explorador del espacio latente

Demostración académica con Fashion-MNIST. Una muestra sintética no garantiza originalidad ni privacidad.

Latente X

1.50

Latente Y

-1.50



Punto latente (1.5, -1.5)

INF-8239 · Explorador del espacio latente

Demostración académica con Fashion-MNIST. Una muestra sintética no garantiza originalidad ni privacidad.

Latente X

0.00

Latente Y

1.50



Punto latente (0.0, 1.5)

INF-8239 · Explorador del espacio latente

Demostración académica con Fashion-MNIST. Una muestra sintética no garantiza originalidad ni privacidad.

Latente X

-1.50

Latente Y

1.50



Punto latente (-1.5, 1.5)

INF-8239 · Explorador del espacio latente

Demostración académica con Fashion-MNIST. Una muestra sintética no garantiza originalidad ni privacidad.

Latente X

-3.00

Latente Y

-3.00



Punto latente (-3.0, -3.0)

INF-8239 · Explorador del espacio latente

Demostración académica con Fashion-MNIST. Una muestra sintética no garantiza originalidad ni privacidad.

Latente X

3.00

-3.00 3.00

Latente Y

3.00



Punto latente (3.0, 3.0)

INF-8239 · Explorador del espacio latente

Demostración académica con Fashion-MNIST. Una muestra sintética no garantiza originalidad ni privacidad.

Latente X

-3.00

Latente Y

3.00



Punto latente (-3.0, 3.0)

INF-8239 · Explorador del espacio latente

Demostración académica con Fashion-MNIST. Una muestra sintética no garantiza originalidad ni privacidad.

Latente X

3.00

Latente Y

-3.00



Punto latente (3.0, -3.0)

INF-8239 · Explorador del espacio latente

Demostración académica con Fashion-MNIST. Una muestra sintética no garantiza originalidad ni privacidad.

Latente X

-1.50

Latente Y

-3.00

-3.00 3.00



Punto latente (-1.5, -3.0)



Al observar las coordenadas exploradas, podemos mapear visualmente cómo el VAE organizó las categorías de Fashion-MNIST en su espacio de 2 dimensiones:

- **Calzado (Y negativo):** En la zona de Y = -3.0, el modelo genera zapatos. Al mover el Latente X desde -3.0 hasta 0.0, la silueta transita fluidamente de una bota a una zapatilla deportiva.
- **Pantalones (X positivo, Y neutral):** En las coordenadas (1.5, 0.0) y (3.0, 0.0), el modelo define claramente siluetas de pantalones.
- **Prendas superiores (Y positivo):** En las regiones de Y = 1.5 a 1.8 con X negativo o neutral, aparecen variaciones de camisetas y suéteres de manga larga.

Además, la interfaz expone correctamente la advertencia ética requerida sobre el uso de modelos generativos: *"Una muestra sintética no garantiza originalidad ni privacidad"*.